

Reflection Groups and Coxeter Groups

Labix

January 16, 2026

Contents

1	Reflection Groups	3
1.1		5
1.2	Coxeter Groups	6
1.3	Decomposing Root Systems into Irreducibles	6
1.4	Dynkin Diagrams and Cartan Matrices	7
1.5	Classification of Irreducible Root Systems	7

1 Reflection Groups

Definition 1.0.1 (Reflections)

Let E be a Euclidean space. Let $T : E \rightarrow E$ be a linear map. We say that T is a reflection if

$$\{v \in E \mid T(v) = v\}$$

is an $(n - 1)$ -dimensional subspace of E .

Definition 1.0.2 (The Reflection Associated to a Vector)

Let E be a Euclidean space. Let $x \in E$. Define the reflection associated to E to be the map $S_x : E \rightarrow E$ given by

$$S_x(z) = z - 2 \frac{\langle x, z \rangle}{\langle x, x \rangle} x$$

Lemma 1.0.3

Let E be a Euclidean space. Let $x \in E$. Then the following are true.

- S_x is a reflection.
- $S_x = S_{\lambda x}$ for any $\lambda \in \mathbb{R} \setminus \{0\}$.

Definition 1.0.4 (Root of a Reflection)

Let E be a Euclidean space. Let $T : E \rightarrow E$ be a reflection. Let $x \in E$. We say that x is a root of T if we have $T = S_x$.

Proposition 1.0.5

Let E be a Euclidean space. Let $T : E \rightarrow E$ be a reflection. Then the following are true.

- T admits a root.
- The set of roots of T is given by

$$\{v \in E \mid T(v) = v\}^\perp$$

Lemma 1.0.6

Let E be a Euclidean space. Let $T : E \rightarrow E$ be a reflection. Then $T^2 = \text{id}_E$.

Definition 1.0.7 (Finite Reflection Groups)

Let E be a Euclidean space. Let $G < O(E)$. We say that G is a finite reflection group if the following are true.

- G is finite.
- G is generated by reflections.

Example 1.0.8

Consider the representation

$$D_{2n} \cong \left\langle \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right\rangle \leq O(\mathbb{R}^2)$$

Then D_{2n} is a finite reflection group.

Example 1.0.9

Consider S_n as a subgroup of $O(\mathbb{R}^n)$ via the injection

$$\sigma \mapsto (e_i \mapsto e_{\sigma(i)})$$

Then S_n is a finite reflection group.

Proof

Indeed, S_n is generated by transpositions, and transpositions are reflections since for a transposition $(i \ j)$, the corresponding element in $O(\mathbb{R}^n)$ fixes e_k for $k \neq i, j$ and transposes $e_i - e_j$ with $e_j - e_i$. ■

Definition 1.0.10 (Root Systems of a Finite Reflection Group)

Let E be a Euclidean space. Let G be a finite reflection group of E . Define the root system of G to be

$$\Phi_G = \{x \in E \mid S_x \in G \text{ and } \|x\| = 1\}$$

Definition 1.0.11 (Abstract Root Systems)

Let E be a Euclidean space. Let $\Phi \subseteq E$ be a subset. We say that Φ is a root system if the following are true.

- Φ is finite and $0 \notin \Phi$.
- If $x \in \Phi$ and $\lambda x \in \Phi$, then $\lambda = \pm 1$.
- If $x, y \in \Phi$, then $S_x(y) \in \Phi$.

Note: Our definition of root systems does not include the requirement that Φ must span E , and that $2\frac{\langle x, y \rangle}{\langle y, y \rangle} \in \mathbb{Z}$.

Lemma 1.0.12

Let E be a Euclidean space. Let G be a finite reflection group of E . Then Φ_G is a root system.

Definition 1.0.13 (The Finite Reflection Group of a Root System)

Let E be a Euclidean space. Let $\Phi \subseteq E$ be a root system. Define the finite reflection group associated to Φ to be

$$G_\Phi = \langle S_x \mid x \in \Phi \rangle$$

Lemma 1.0.14

Let E be a Euclidean space. Let $\Phi \subseteq E$ be a root system. Then G_Φ is a finite reflection group.

Note: There can be multiple different root systems that give rise to the same finite reflection group. Note: $\{S_x \mid x \in \Phi_G\}$ might not be the complete set of reflections of G .

Definition 1.0.15 (Simple Systems)

Let E be a Euclidean space. Let Φ be a root system. Let $\Delta \subseteq \Phi$ be a subset. We say that Δ is a simple system for Φ if the following are true.

- Δ is a basis for $\mathbb{R}\langle\Phi\rangle$.
- For any $r \in \Phi$, there is a decomposition

$$r = \sum_{b \in \Delta} k_b \cdot b$$

where $k_b \in \mathbb{R}$ are either all positive or all negative.

Lemma 1.0.16

Let E be a Euclidean space. Let Φ be a root system. Let Δ be a simple system of Φ . Let $x, y \in \Phi$ be distinct. Then we have

$$\langle x, y \rangle < 0$$

Lemma 1.0.17

Let E be a Euclidean space. Let Φ be a root system. Let Δ be a simple system of Φ . Then we have

$$G_\Phi = \langle S_x \mid x \in \Delta \rangle$$

Corollary 1.0.18

Let E be a Euclidean space. Let Φ be a root system. Let Δ be a simple system of Φ . Then for any $x \in \Phi$, there exists $y \in \Phi$ and $T \in G_\Phi$ such that $T(y) = x$.

2 Crystallographic Root Systems

2.1 The Integrality Condition

Definition 2.1.1 (Crystallographic Root Systems)

Let E be a Euclidean space. Let Φ be a root system. We say that Φ is crystallographic if the following are true.

- Φ spans E .
- For any $x, y \in \Phi$, we have $2\frac{\langle x, y \rangle}{\langle y, y \rangle} \in \mathbb{Z}$.

Example 2.1.2

Let $\{e_1, \dots, e_{n+1}\}$ be the standard basis vectors of $\mathbb{R}^{n+1}$. Let E be the inner product space

$$E = \{(c_1, \dots, c_{n+1}) \subseteq \mathbb{R}^{n+1} \mid c_1 + \dots + c_n = 0\} \cong \mathbb{R}^n$$

equipped with the standard inner product of $\mathbb{R}^{n+1}$. Then the set

$$R = \{e_i - e_j \mid 1 \leq i \neq j \leq n+1\}$$

is a root system of E .

Lemma 2.1.3

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a root system of E . Let $\alpha, \beta \in R$. Then the following are true.

- The value of $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$ must lie in the set $\{0, 1, 2, 3\}$
- The angle between α and β must lie in the set

$$\left\{ \frac{\pi}{2}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{6}, \frac{5\pi}{6} \right\}$$

We can draw a table of the possible values of $\langle \alpha, \beta \rangle$ and the corresponding angle between α and β :

$\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$	$\langle \alpha, \beta \rangle$	$\langle \beta, \alpha \rangle$	θ
0	0	0	$\frac{\pi}{2}$
1	1	1	$\frac{\pi}{3}$
	-1	-1	$\frac{2\pi}{3}$
2	1	2	$\frac{\pi}{4}$
	-1	-2	$\frac{3\pi}{4}$
3	1	3	$\frac{\pi}{6}$
	-1	-3	$\frac{5\pi}{6}$

Here we assume $\|\alpha\| \leq \|\beta\|$ so we exclude for instance the case $\langle \alpha, \beta \rangle = 2$ and $\langle \beta, \alpha \rangle = 1$.

Example 2.1.4

The only root system in $\mathbb{R}^1$ is given by $R = \{\alpha, -\alpha\}$ for any $\alpha \in \mathbb{R}$.

Example 2.1.5

There are exactly four possible root systems in $\mathbb{R}^2$. Let $\alpha, \beta \in \mathbb{R}^2$ be linearly independent such that $\|\alpha\| \leq \|\beta\|$. Let θ be the angle between α and β .

- $A_1 \times A_1 = \{\pm\alpha, \pm\beta\}$ and $\theta = \frac{\pi}{2}$.
- $A_2 = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta)\}$ and $\theta = \frac{2\pi}{3}$.
- $B_2 = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta), \pm(2\alpha + \beta)\}$ and $\theta = \frac{3\pi}{4}$.

$$\bullet G_2 = \{\pm\alpha, \pm\beta, \pm(\alpha + \beta), \pm(2\alpha + \beta), \pm(3\alpha + \beta), \pm(3\alpha + 2\beta)\}$$

Definition 2.1.6 (Base of a Crystallographic Root System)

Let E be a Euclidean space. Let Φ be a root system. Let $B \subseteq \Phi$ be a subset. We say that B is a base for Φ if the following are true.

- B is a basis for $\mathbb{R}\langle\Phi\rangle = E$.
- For any $r \in \Phi$, there is a decomposition

$$r = \sum_{b \in \Delta} k_b \cdot b$$

where $k_b \in \mathbb{Z}$ are either all positive or all negative.

Proposition 2.1.7

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let Φ be a crystallographic root system of E . Then Φ admits a base.

2.2 Decomposing Root Systems into Irreducibles

Definition 2.2.1 (Irreducible Root Systems)

Let $(E, (-, -))$ be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E . We say that R is reducible if there exists two disjoint crystallographic root systems R_1 and R_2 of E such that $(r_1, r_2) = 0$ for $r_1 \in R_1$ and $r_2 \in R_2$. Otherwise, we say that R is irreducible.

Proposition 2.2.2

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E . Then the following are true.

- There is a decomposition

$$R = R_1 \amalg \cdots \amalg R_k$$

where each R_i is an irreducible crystallographic root system of $E_i = \mathbb{R}\langle R_i \rangle$

- There is a decomposition

$$E = E_1 \oplus \cdots \oplus E_k$$

where E_i is orthogonal to E_j for $i \neq j$.

Definition 2.2.3 (Isomorphic Root Systems)

Let E, F be a finite dimensional inner product spaces over $\mathbb{R}$. Let R and S be crystallographic root systems of E and F respectively. We say that R and S are isomorphic if there exists an isomorphism

$$\phi : E \rightarrow F$$

of vector spaces such that the following are true.

- $\phi(R) = S$
- For all $a, b \in R$, $\langle \phi(a), \phi(b) \rangle = \langle a, b \rangle$.

Example 2.2.4

The following are true.

- $A_1 \times A_1 = \{\pm\alpha, \pm\beta\}$ is reducible into the product of A_1 and A_1 .
- A_2, B_2, G_2 are irreducible.

2.3 Dynkin Diagrams and Cartan Matrices

Definition 2.3.1 (The Dynkin Diagram of a Root System)

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E . Let B be a base of R . Define a graph $\Delta(R)$ as follows.

- There is one vertex v_b for each $b \in B$
- For any two vertices v_a and v_b , there are $d_{a,b} = \langle a, b \rangle \langle b, a \rangle$ number of undirected edges between v_a and v_b .

Proposition 2.3.2

Let E, F be a finite dimensional inner product spaces over $\mathbb{R}$. Let R and S be crystallographic root systems of E and F respectively. Then $R \cong S$ if and only if $\Delta(R) \cong \Delta(S)$.

Proposition 2.3.3

Let E be a finite dimensional inner product space. Let R be a crystallographic root system of E . Then R is irreducible if and only if $\Delta(R)$ is a connected graph.

Definition 2.3.4 (The Cartan Matrix)

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E . Let $B = \{\alpha_1, \dots, \alpha_r\}$ be a base of R . Define the Cartan matrix of R to be

$$\begin{pmatrix} \langle \alpha_1, \alpha_1 \rangle & \cdots & \langle \alpha_1, \alpha_r \rangle \\ \vdots & \ddots & \vdots \\ \langle \alpha_r, \alpha_1 \rangle & \cdots & \langle \alpha_r, \alpha_r \rangle \end{pmatrix}$$

Proposition 2.3.5

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E . Then the Cartan matrix of R is unique up to reordering.

2.4 Classification of Irreducible Root Systems

Theorem 2.4.1 (Classification of Irreducible Root Systems)

Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be an irreducible crystallographic root system of E . Then $\Delta(R)$ is isomorphic to one of the following:

- A_n for $n \geq 1$.
- B_n for $n \geq 2$.
- C_n for $n \geq 3$.
- D_n for $n \geq 4$.
- G_2, F_4, E_6, E_7, E_8 .

Conversely, any of the above types can be realized as the Dynkin diagram of some irreducible root system.

2.5 The Weyl Group of a Root System

Definition 2.5.1 (Weyl Group of a Root System)

Let E be a Euclidean space. Let Φ be a crystallographic root system. Define the Weyl group of Φ to be

$$W(\Phi) = G_\Phi = \langle S_x \mid x \in \Phi \rangle$$

3 Coxeter Groups

3.1 Coxeter Groups

Definition 3.1.1 (Coxeter Groups)

Let G be a group. We say that G is a Coxeter group if G admits a presentation

$$G \cong \langle r_1, \dots, r_n \mid (r_i r_j)^{m_{ij}} \rangle$$

where $m_{ii} = 1$ and $m_{ij} \geq 2$ or ∞ for $i \neq j$.

Proposition 3.1.2

Let E be a Euclidean space. Let G be a finite reflection group of E . Then G is a Coxeter group.